

WSM-WWB Bridge user guide

Moving wireless-mic coordination data between **Shure Wireless Workbench** and **Sennheiser Wireless Systems Manager**, plus generic CSV.

Two things before you use it on a show

Every parser was reverse-engineered

Neither vendor publishes full schemas for most of these files. Everything this tool understands was worked out by inspecting real exports and cross-checking two independent formats from the same project against each other.

Verify the output against your own WWB/WSM versions before relying on it for a live show. The formats were read from **WWB 7.8.2.63** and **WSM 4.9.0.13**; a different version may differ.

There is a successor

RFutils merged this tool with MicWizard and `pmse-to-wwb` into one suite. This repo stays a standalone, MIT-licensed, releasable desktop tool — but RFutils is where new feature work goes.

The workflow

1. **Open File...** — the format is auto-detected.
2. **Check the preview table** — Name, Frequency, Zone, Group, Channel, Type, Manufacturer, Notes.
3. **Pick an export format and Save As...**

If a file isn't recognised, you get a **column-mapping dialog** instead of an error. That's the designed fallback, not a failure.

What it can read

Source	Format
Shure WWB	<code>.shw</code> (Show) and <code>.cws</code> (Coordination Workspace) native XML
	Coordination report CSV (the printable/exportable one)
	A bare frequency list
Sennheiser WSM	<code>.wsm</code> native project file
	HTML "Coordination Report"
	"Frequencies/Bands" CSV export
Anything else	via the column-mapping dialog

Which WWB file you open changes what you get

- A `.shw` carries the **deployed device inventory** — what's actually on the gear. It also embeds workspace data, and the reader **prefers the device inventory** when both are present.
- A `.cws` carries the **candidate frequency pool** across all RF zones — the coordination engine's options, not the deployment.

Those are very different lists. If a file yields far more channels than you expected, you're looking at the candidate pool.

Only the WWB report knows primary vs backup

Neither `.shw` nor `.cws` exposes a primary/backup flag at the XML level — **only the printed coordination report does**. So if you need "what am I actually using" versus "what's in the spare pool", **open the report CSV**, not the native file.

(The report was verified against a real 291-channel export: 44 primary and 247 backup across 3 zones, matching the file's own section counts exactly.)

WSM `.wsm` has two frequency fields and they disagree

A real coordinated project had `CurrentFrequency` **sitting at the receiver's default** — not reflecting the coordination result at all — while `AllocatedFrequency` matched the real result. **The tool reads `AllocatedFrequency`**. If WSM shows you one number and this tool shows another, that's the field difference, and the tool is reading the coordinated one.

What it can write, and which to choose

Target	Use	Why
WWB frequency list	The safe default for WSM → WWB	The one format Shure documents as importable
WWB inventory CSV	round-tripping with this tool	Not a real WWB format — best-effort flat CSV
WSM "Frequencies/Bands" CSV	feeding a candidate pool into WSM	Not a coordinated channel list — see below
Generic CSV	anything else	

Writing `.shw` / `.cws` is deliberately not implemented

Those files have many interdependent sections — compatibility profiles, band planning, zone matrices — well beyond channel data. Generating one from scratch risks producing a file WWB can't open cleanly. **Reading them is safe; the safe way to get data *into* WWB is the frequency list import.**

The WSM CSV export does not populate named channels

WSM's "Frequencies/Bands" CSV feeds a **candidate frequency pool** into WSM. You then run WSM's own **Start Coordination** and drag-allocate frequencies onto device channels yourself.

It will not write coordinated frequencies onto your named channels. If that's what you wanted, this format won't do it — there is no export that does, because WSM doesn't import one.

Frequencies in that file are in **kHz**, and the real schema is lowercase and semicolon-delimited — which **differs from Sennheiser's own documentation**. The real file won.

Troubleshooting

Symptom	Cause
A column-mapping dialog appeared	The file wasn't recognised. Expected behaviour, not an error (The workflow).
Far more channels than expected from a WWB file	You opened a <code>.cws</code> , or a <code>.shw</code> whose candidate pool was read — that's the whole candidate pool (What it can read).
No primary/backup distinction	Only the WWB coordination report carries it (What it can read).
Frequencies differ from what WSM shows	WSM's <code>CurrentFrequency</code> is often the receiver default; the tool reads <code>AllocatedFrequency</code> (What it can read).
A European-locale file parsed correctly	Expected — a comma with no dot is read as a decimal point.
Imported into WSM and nothing landed on my channels	The CSV is a candidate pool. Run Start Coordination and allocate (What it can write, and which to choose).
Can't export a <code>.shw</code>	Intentional — write-back isn't implemented (What it can write, and which to choose).
Output looks wrong against my WWB/WSM version	Formats were read from WWB 7.8.2.63 / WSM 4.9.0.13. Verify before a show (Two things before you use it on a show).
macOS says the app is damaged	Unsigned build; see the README's Gatekeeper section.

See also

- [API.md](#) — the module surface, the channel model, format detection
- [DEVELOPING.md](#) — extending the parsers
- [README Format notes](#) — what each format actually contains